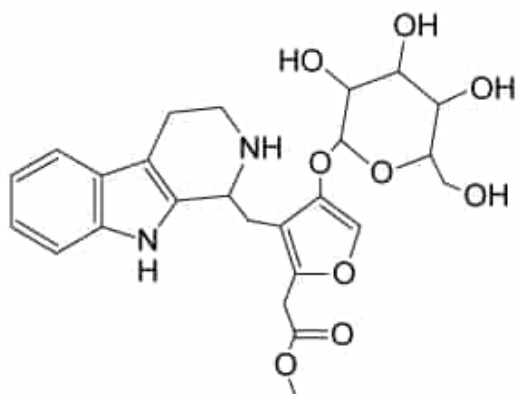


Which of the following amino acids would most likely be used as a precursor in the synthesis of Compound X, shown below?



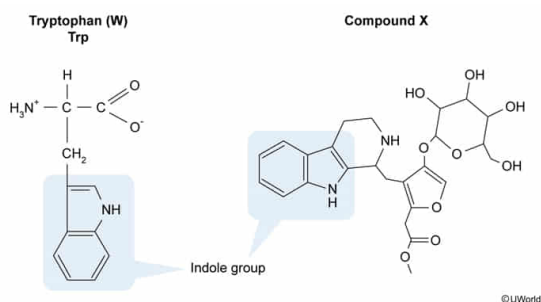
Compound X

- ☐ A. Pro [7%]
☒ B. Trp [71%]
☐ C. His [15%]
☐ D. Tyr [5%]

Correct

72% answered correctly

Explanation:



Amino acids are distinguished by the **chemical groups** on their side chains, which confer different chemical and biological properties on those amino acids. Many amino acids have relatively **simple functional groups**, including alkyl chains (Ala, Leu, Ile, Val), alcohols (Ser, Thr), carboxylic acids (Glu, Asp), amides (Gln, Asn), amines (Lys), and thiols (Cys). A few amino acids have more **complex side chains**, such as **imidazole** (His) and **indole** (Trp) groups.

The indole structure in the tryptophan (Trp) side chain is incorporated into several biologically important molecules, such as serotonin and melatonin. It is present in Compound X, and therefore tryptophan is the most likely amino acid precursor in the synthesis of the compound.

(Choice A) The side chain of **proline** is a cyclic alkyl group that bonds to the nitrogen of the amino acid backbone in a five-membered ring. It is not present in Compound X.

(Choice C) The side group of **histidine** is imidazole, a five-membered aromatic ring containing two nitrogen atoms. Compound X does not include imidazole.

(Choice D) The side group of **tyrosine** is phenol. Compound X does not contain phenol.

Educational objective:

Each amino acid has a unique side chain that confers specific chemical and biological properties on that amino acid. Side chains may have relatively simple functional groups, such as alkyl chains, alcohols, and carboxylic acids, or they may have more complex groups, such as indole or imidazole.

Time spent:0 sec

Subject:
Organic Chemistry

QID:401476

System:
Functional Groups and Their Reactions